# Bright Motors Car Sales Analysis (BRIGHTLEARN)

**Purpose:** Business Insights for a New Head of Sales using Historical Car Sales Data from Bright Motors

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## 1. INTRODUCTION & INSTRUCTIONS

### 1.1 INTRODUCTION

You have been provided with a dataset titled **"Bright Car Sales"**, which captures daily transactional and pricing information for vehicles sold by Bright Motors.

Bright Motors has recently appointed a **new Head of Sales**, whose mission is to expand the dealership network, improve sales performance, and optimize inventory. Your role, as a **Junior Data Analyst**, is to extract actionable insights from the historical car sales data and prepare a presentation that will help guide future sales and marketing strategies.

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## 2. OBJECTIVE

Use your data analytics, SQL, and data visualization skills to help Bright Motors understand:

- Which car makes and models generate the most revenue
- The relationship between price, mileage, and year of manufacture
- Which regions or locations have the highest sales volumes
- Emerging trends in customer purchasing preferences
- Recommendations to increase dealership profitability and efficiency

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## 3. TOOLS ALLOWED

You may use any of the following tools (or equivalents):

**Coding / SQL Platforms:**

- Snowflake

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- Microsoft SQL Server
- Databricks
- Google BigQuery
- MySQL Workbench

**Data Visualization:**

- Microsoft Excel
- Power BI
- Google Sheets
- Google Looker Studio

**Presentation / Reporting:**

- Microsoft PowerPoint
- Canva

**Project Planning:**

- Miro
- Figma

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## 4. TASKS (Detailed)

### Task 1 — Planning & Architecture (Miro or equivalent)

Create a Miro board (or any whiteboard tool) that outlines:

1. **Data Flow / Architecture Diagram** showing:
    - **Source**: car sales dataset (CSV/Excel file)
    - **ETL Pipeline**: data cleaning (e.g., removing duplicates, handling missing values, currency formatting)
    - **Storage**: Snowflake or chosen SQL database
    - **Analysis Layer**: SQL and visualization tools (Excel / Power BI)
    - **Presentation Layer**: PowerPoint / Canva
2. **Key Insights to Deliver:**
    - Revenue by car make and model
    - Sales distribution by year and fuel type
    - Regional performance (city or province)

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- Average selling price trends over time

3. **Key Calculations:**
   - `Total_Revenue = Selling_Price * Units_Sold`
   - `Profit_Margin = (Selling_Price - Cost_Price) / Selling_Price * 100`
   - Grouping by `Make`, `Model`, `Year`, `Region`, and `Fuel_Type`

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## Task 2 — Data Processing in Snowflake (or chosen SQL platform)

**Steps:**

1. Convert the provided Excel file into a CSV file.

2. Load the CSV file into Snowflake (or equivalent).

3. Perform data cleaning and transformations:

   - Convert text-based prices (e.g., '15,000') to numeric format.
   - Create a new column `total_revenue = selling_price * units_sold`.
   - Calculate `profit_margin` and categorize cars by performance tiers (e.g., *High Margin*, *Medium Margin*, *Low Margin*).
   - Group transactions by time periods (month, quarter, or year).

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## Task 3 — Data Analysis & Visualization (Excel / Power BI / Google Looker Studio)

- After transforming your data in Snowflake, export the processed dataset to Excel or connect Power BI directly to your database.
- Use slicers for region, fuel type, and year to allow interactivity.

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## Task 4 — Presentation to the Head of Sales

- Create a professional presentation that summarizes findings and recommendations.

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# 5. SUBMISSION GUIDELINES

**Required files:**

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- Miro Plan / Architecture Diagram (image or link)
- Processed Dataset Spreadsheet (`car_sales_processed.xlsx`)
- Presentation File (`BrightMotors_Presentation.pdf`)
- SQL Script (`car_sales_queries.sql`)

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## 6. TIPS & NOTES

- Ensure all prices are converted to numeric formats (remove commas or currency symbols).
- Handle missing or inconsistent values carefully.
- Use clear and consistent column names in Snowflake.
- Include short insight notes in your dashboards.
- Your presentation should be visually appealing and concise.

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**End of Case Study**

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